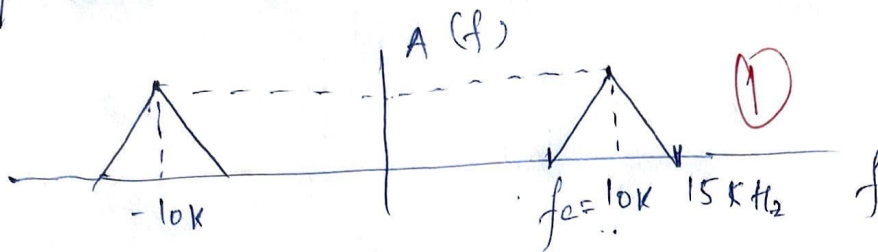


1.1 At C: $(m_1(t) + 2m_2(t) \cos(20,000\pi t)) (2\cos(40,000\pi t))$ (1)

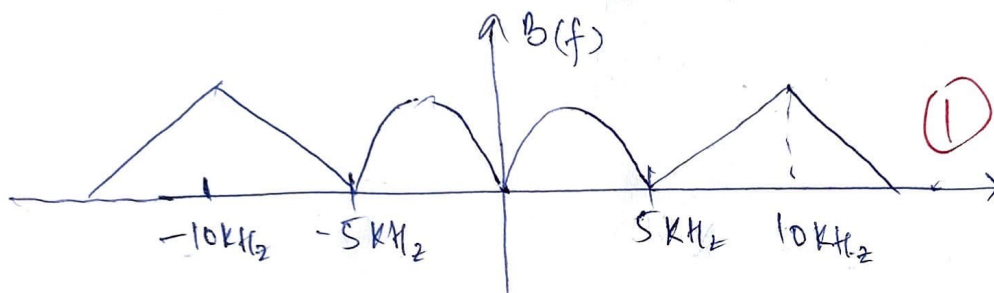
Freq. domain representation at A.



Freq. domain representation at B:

$$m_1(t) + 2m_2(t) \cos(20,000\pi t) \leftrightarrow$$

$$M_1(f) + 2M_2(f) * \frac{1}{2} (\delta(f+f_c) + \delta(f-f_c))$$



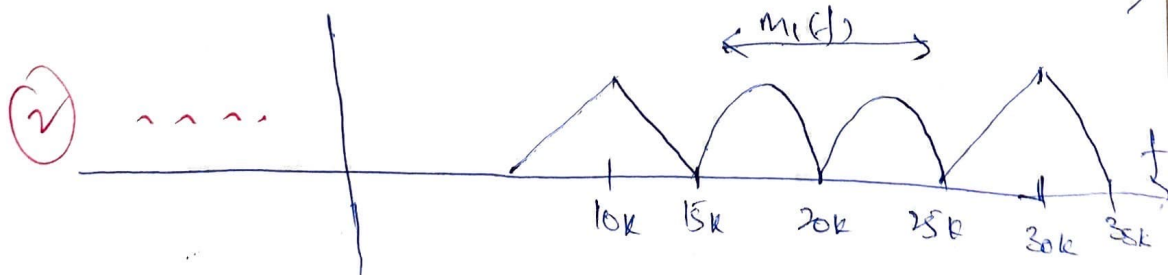
Freq. domain at C:

$$m_1(t) \cdot 2\cos(40,000\pi t) + 4m_2(t) \cos(20,000\pi t) \cos(40,000\pi t)$$

$$= 2m_1(t) \cos(40,000\pi t) + 4m_2(t) (\cos(20,000\pi t) \cos(40,000\pi t))$$

$$\leftrightarrow M_1(f) * (\delta(f-20k) + \delta(f+20k))$$

$$+ M_2(f) * (\delta(f-10k) + \delta(f+10k) + \delta(f-30k) + \delta(f+30k))$$



(1.2) $\hat{x}(t) = x(t) * \frac{1}{\pi t}$

$\updownarrow$

$\hat{x}(f) = -j \operatorname{sgn}(f) X(f)$

Now let Hilbert transform of $\hat{x}(t) = \hat{\hat{x}}(t)$

$\hat{\hat{x}}(t) = \hat{x}(t) * \frac{1}{\pi t}$

$\updownarrow$

$\hat{\hat{x}}(f) = -j \operatorname{sgn}(f) \hat{x}(f)$

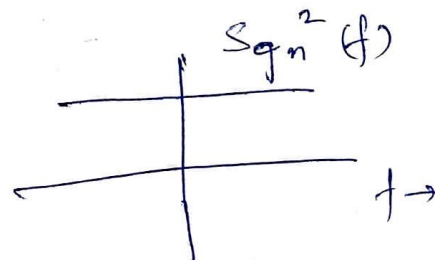
$= -j \operatorname{sgn}(f) [-j \operatorname{sgn}(f) X(f)]$

$= - \operatorname{sgn}^2(f) X(f)$

$= -X(f)$

$\hat{\hat{x}}(t) = -x(t)$

-(2)



Any other method is also fine.

(1.3) $m(t)$.

$$B = 10 \text{ kHz}$$

$$m_p = 1 \text{ V}$$

$$\therefore \frac{\Delta f}{m_p} = 100 \text{ Hz/V} \quad \Rightarrow \quad \Delta f = 100 \text{ Hz}$$

$$B_{fm} = 2B (\beta + 1)$$

$$\beta = \frac{\Delta f}{B} = \frac{100}{10 \text{ K}} = 10 \times 10^{-3}$$

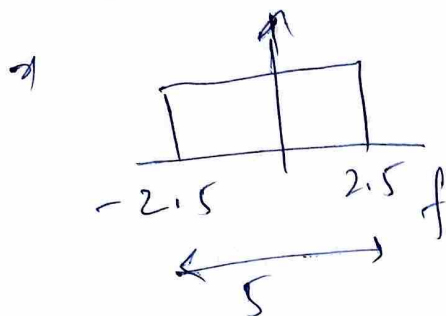
$$\rightarrow B_{fm} = 2 \times 10 \times 10^3 (1 + 0.001)$$

$$= 20 \times 10^3 \times 1.01$$

$$= 20.2 \text{ kHz.}$$

(3)

(2.1) $g(t) = \text{Sinc}(.5\pi t)$

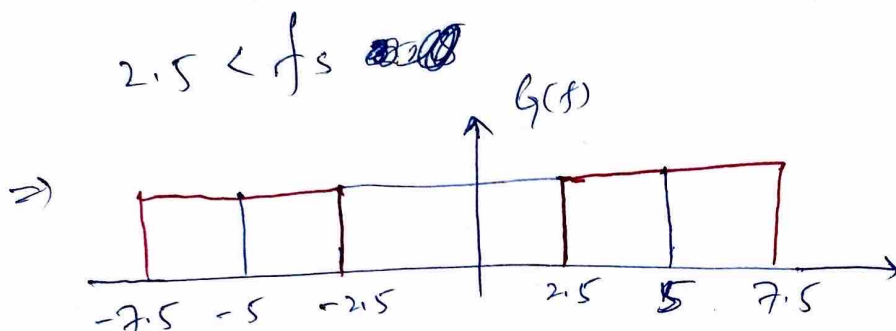


$$B = 2.5 \text{ Hz.}$$

$$\text{Nyquist } f_s = 2B = 5 \text{ Hz}$$

(i) Sampling rate = 2.5 Hz.

$$2.5 < f_s$$

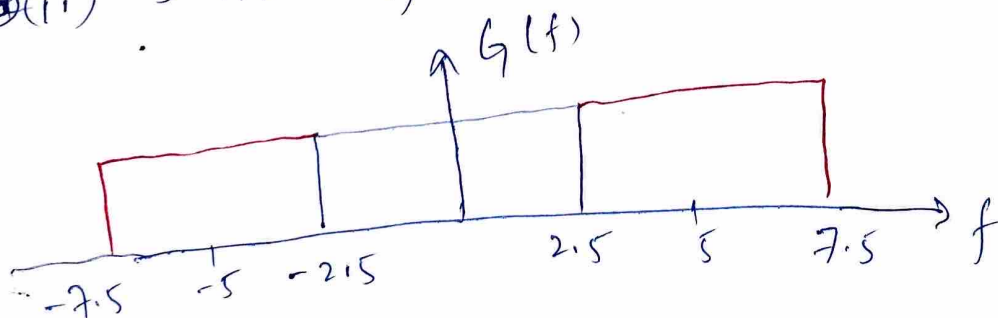


Here there is overlapping of the samples because Nyquist criteria is not fulfilled.

Hence we can not recover the $g(t)$ from sampled signal.

--- (3)

(ii) 5 Hz. = f_s .



Here Nyquist criteria is fulfilled as there is no overlap.

→ $g(t)$ can be recovered..

2.2

$$m_p = 4V$$

$$L = 256 \rightarrow n = 8 \text{ bit}$$

$$F_m = 15 \text{ kHz} \rightarrow \text{nyquist rate} = 30 \text{ kHz}$$

$$F_s = \text{~~0.0~~ } 37.5 \text{ kHz} \quad (1.25 \times 30)$$

$$\begin{aligned} \nearrow \text{Max. Quantization Error} &= \frac{\Delta V}{2} = \frac{m_p}{L} = \frac{4}{256} \\ &= 0.015625 \text{ V} \end{aligned}$$

$$\text{Bit rate} = n f_s$$

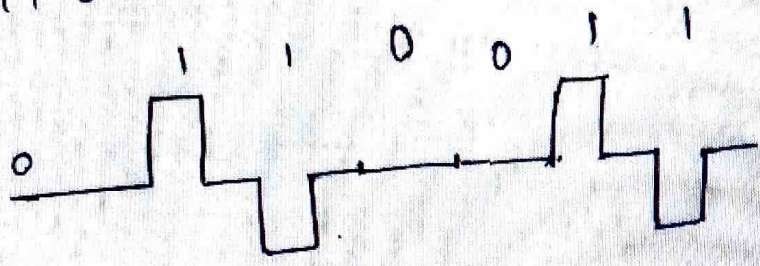
$$= 8 \times 37.5$$

$$= 300 \text{ Kbps.} \dots$$

2.3

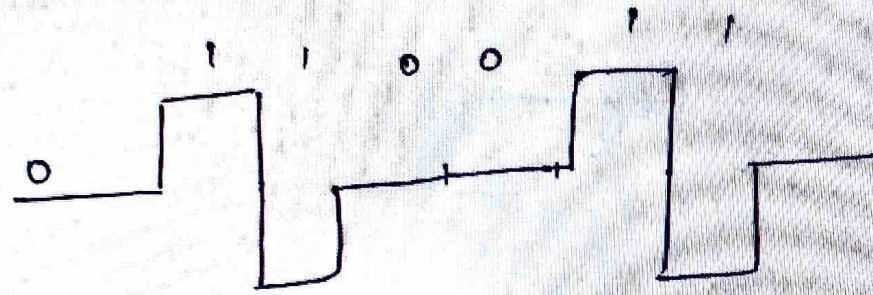
0 1 1 0 0 1 1

RZ:



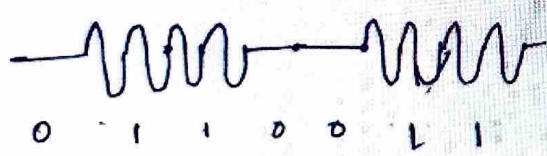
0.5

NRZ:



(a) ASK:

RZ:



Bit duration: 1s

$$f_c = 2Hz$$

→ in 1 bit we have 2 cycles.

0.75

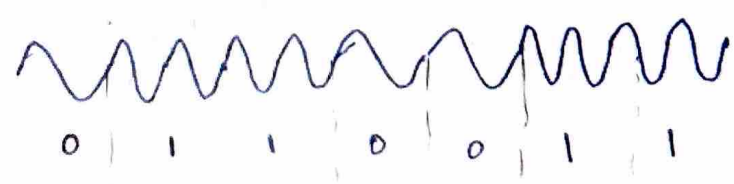
NRZ: Same

(b) FSK with $1Hz$ & $2Hz$.

$1Hz \Rightarrow 1$ cycle

$2Hz \Rightarrow 2$ cycles

} need to be computed else-0-



0.75

3.1

$$\phi(t) = \begin{cases} 1 & t=0 \\ 0 & t = \pm nT_b \end{cases} \left(\frac{1}{T_b} \right) \left(\frac{1}{R_b} \right) \dots \textcircled{1}$$

→ Raised cosine : practically realizable. ①

Sine pulse → Infinite frequency response.

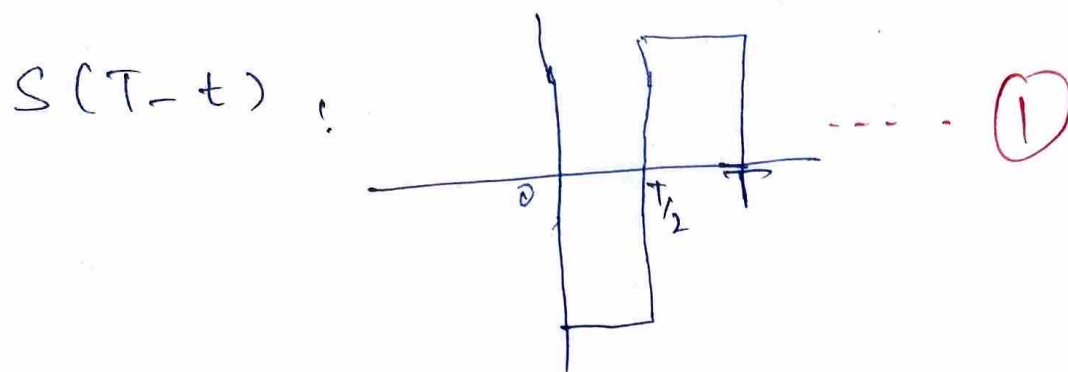
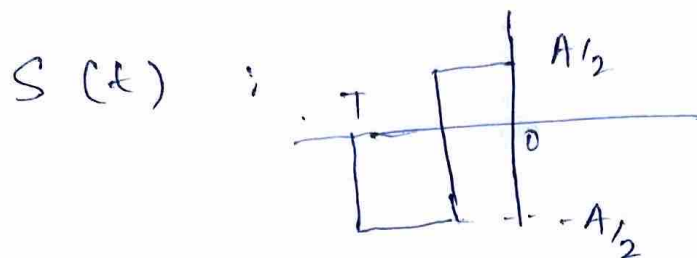
Raised cosine : Adjustable B.W w.r.t to roll-off factor.

: decays rapidly ($\frac{1}{t^3}$)

→ Insensitive to deviations of R_b .
Sampling rate,

3.2

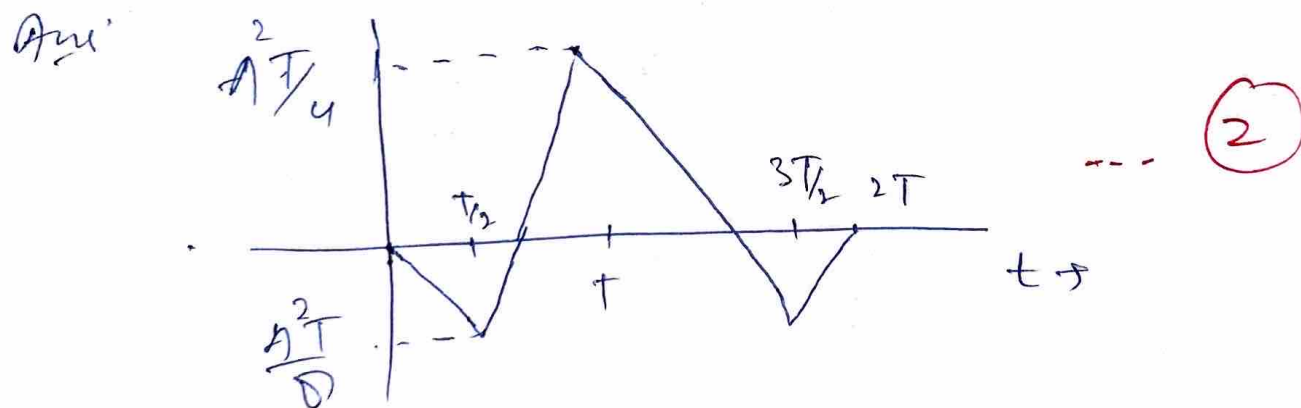
Matched filter $S(T-t)$



e/p of matched filter $s(t) * S(T-t)$

$$= \int_0^T s(\tau) s(T-\tau) d\tau$$

Need part-wise calculation & steps involved



$$3.3. \quad m_x(t) = 3t \quad C_x(t_1, t_2) = 9 \exp\{-2|t_1 - t_2|\}$$

(a) $P(X(3) < 6)$: This requires computation of mean & variance of a single r.v. $X(3)$.

$$X(3) \sim N(m_x(3), C_x(3,3))$$

$$= N(9, 9 \cdot \exp(0)) = N(9, 9) \quad \dots (0.5)$$

$$\Rightarrow X(3) \sim \frac{1}{(\sqrt{2\pi \cdot 9})} \cdot \exp\left\{-\frac{(x-9)^2}{2(9)}\right\}$$

$$\sim \frac{1}{3\sqrt{2\pi}} \exp\left\{-\frac{(x-9)^2}{18}\right\}$$

$$\Rightarrow P(X(3) < 6) = \int_{-\infty}^6 \frac{1}{3\sqrt{2\pi}} \exp\left\{-\frac{(x-9)^2}{18}\right\} dx \quad \dots (1)$$

(cdf)

(b) Given random variable $X(1) + X(2)$

$$E[X(1) + X(2)] = 3(1) + 3(2) = 9$$

$$\begin{aligned} \text{Var}[X(1) + X(2)] &= C_x(1,1) + C_x(2,2) + C_x(2,1) \\ &\quad + C_x(1,2) \\ &= 9(2 + 2e^{-2}) \end{aligned}$$

$$\Rightarrow P(X(1) + X(2) > 2) = \int_2^{\infty} \frac{1}{\sqrt{2\pi(9(2+2e^{-2}))}} \cdot \exp\left\{-\frac{(x(1)+x(2)-9)^2}{2 \cdot 9(2+2e^{-2})}\right\} dx$$

1.5

$X_n \sim N(0, \sigma^2)$, i.i.d.

3.4 $S_n = X_1 + X_2 + \dots + X_n$

$\mu_{S_n} = 0$... constant

$$R_X(n_1, n_2) = E[S_{n_1} S_{n_2}] = E[(X_1 + X_2 + \dots + X_{n_1})(X_1 + X_2 + \dots + X_{n_2})]$$

$$= E[X_1^2 + X_2^2 + \dots + ?] = \begin{cases} n\sigma^2 \\ n\sigma^2 \end{cases} \quad \begin{matrix} n_1 = n_2 = n \\ \text{if } n_1 < n_2 \\ \text{or } n_2 < n_1 \\ \downarrow \\ = n \end{matrix}$$

→ Variance is not independent of time n .

1.5

→ not WSS.

Moving average process:

$$Y_n = \frac{X_n + X_{n-1}}{2}$$

$$\mu_Y = \frac{E[X_n] + E[X_{n-1}]}{2} = 0 \rightarrow \text{constant for all } n.$$

$$R_X(n_1, n_2) = E\left[\frac{(X_{n_1} + X_{n_1-1})(X_{n_2} + X_{n_2-1})}{4}\right]$$

$$= E\left[\frac{X_{n_1} X_{n_2}}{4}\right] + E\left[\frac{X_{n_1} X_{n_2-1}}{4}\right] + E\left[\frac{X_{n_2} X_{n_1-1}}{4}\right]$$

$$+ E\left[\frac{X_{n_1-1} X_{n_2-1}}{4}\right] = \begin{cases} \frac{1}{2} \sigma^2 & \text{if } n_1 = n_2 \\ \frac{1}{4} \sigma^2 & \text{if } |n_1 - n_2| = 1 \\ 0 & \text{otherwise} \end{cases}$$

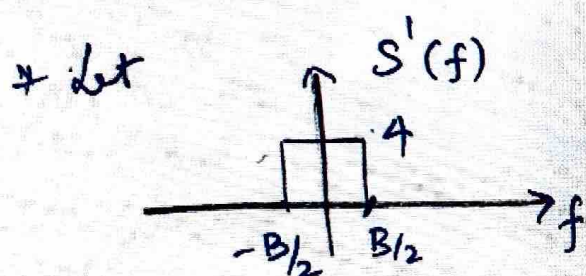
→ It is WSS, depends on $n_1 - n_2$.

3.5. $S_x(f) = \begin{cases} 2 & |f \pm f_0| \leq B/2 \\ 0 & \text{otherwise} \end{cases}$

* Average power : $R_x(\tau) = \int_{-\infty}^{\infty} S_x(f) df$

$$= \int_{f_0 - B/2}^{f_0 + B/2} 2 df + \int_{-(f_0 + B/2)}^{-(f_0 - B/2)} 2 df$$

$$= 2B + 2B = 4B \quad \dots \dots (1)$$



let $S'(f)$ be as given.

$$S_x(f) = \cancel{S'(f)} \cdot S'(f) * \left(\frac{\delta(f-f_0) + \delta(f+f_0)}{2} \right) \quad \dots (2)$$

$$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$$

$$R_x(\tau) = R'(\tau) \cdot \cos 2\pi f_0 \tau$$

From cheatsheet : $S'(f) = \pi(f/B)$

$$\Rightarrow R'(\tau) = B \operatorname{sinc} \left(2\pi \left(\frac{B}{2} \right) \tau \right)$$

$$= B \operatorname{sinc}(\pi B \tau)$$

$$\Rightarrow R_x(\tau) = B \operatorname{sinc}(\pi B \tau) \cos 2\pi f_0 \tau \quad \dots (2)$$

3.6 100 cards

$$P(\text{Ace}) = 3/100 \quad P(\text{Kings}) = 7/100$$

$$P(\text{Tens}) = 34/100$$

$$P(\text{Queens}) = 25/100 \quad P(\text{Jacks}) = 31/100$$

* Information_{Queen} $\log \frac{1}{P(\text{Queen})} = \frac{100}{25} = 4 \log_2(4) = 2$
critical.

* Entropy = $-\sum_{i \in \{\text{Ace}, \dots, \text{Tens}\}} p_i \log_2 p_i = \cancel{\text{Z}}$

$$= -\left(\frac{3}{100} \log_2 \frac{3}{100} + \frac{7}{100} \log_2 \frac{7}{100} + \frac{25}{100} \log_2 \frac{25}{100} \right.$$

$$\left. + \frac{31}{100} \log_2 \frac{31}{100} + \frac{34}{100} \log_2 \frac{34}{100} \right)$$